Contributions

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Introduction 1

1.1 Background

Early identification of individuals at high risk for stroke enables timely intervention. Traditional risk assessment methods rely on clinical expertise and manual scoring systems, which may be subjective and limited in predictive capacity.

This analysis tries to identify key risk factors associated with stroke events and develop a predictive model to predict the likelihood of a stroke. The dataset comprises 9,722 patient records with 18 variables.

1.2 Methodology

The process began with a Data Cleaning phase. Here, the raw data was audited for duplicates that could skew results and missing values, particularly common in medical fields like BMI or smoking history, which were either removed or filled using statistical imputation. We also screened for anomalies or outliers, such as impossible physiological readings, to ensure the model wouldn't learn from "noisy" or incorrect data.

Once the data was cleaned, we moved into the Cluster Analysis to identify the different clusters of patients in the dataset. Following that, we moved into Exploratory Data Analysis (EDA). This involved visualizing distributions to understand the demographics of the patient population and using correlation matrices to see how variables like age, heart disease, and glucose levels interact. A key takeaway from this phase is often identifying class imbalance, as stroke events are typically much rarer than non-events in a general population, a factor that heavily influences how a model is trained.

To move beyond visual trends and into scientific certainty, we performed Statistical Testing. By applying methods such as Chi-Square tests for categorical variables and T-tests for continuous data, we were able to calculate p-values to determine if a correlation was statistically significant. This step filtered out the "noise," ensuring that only true clinical risk factors those with a mathematically proven link to stroke events were prioritized for the final model.

Finally, we synthesized these insights into a Predictive Model, transforming historical data into a forecasting tool. By selecting an appropriate algorithm, such as Logistic Regression for its interpretability or a Random Forest for its ability to handle complex patterns, we trained the system to assign a probability score to new patients. The success of this model likely hinged on optimization metrics like sensitivity or AUC-ROC, ensuring the tool is sensitive enough to catch high-risk cases without an overwhelming number of false alarms.

1.3 Dataset Description

The healthcare dataset contains patient information could be categorized into four main groups:

Table 1.1: Information Types

Category	Variables
Demographic	○ age, ○ age_group, ○ gender, ○ marital_status, ○ employment_type, ○ residence
Clinical	○ has_hypertension, ○ has_heart_disease, ○ glucose_level, ○ high_glucose, ○ bmi_value, ○ bmi_category, ○ smoking_habit, ○ lifestyle_risk
Derived	○ risk_score

The dataset contains variables of different types, continuous and categorical. And the categorical variables can be divided as nominal and ordinal.

Table 1.2: Variable Classification

Category	Variables
Numerical	○ age, ○ glucose_level, ○ bmi_value, ○ risk_score
Ordinal	○ age_group, ○ bmi_category, ○ lifestyle_risk
Nominal	○ gender, ○ has_hypertension, ○ has_heart_disease, ○ marital_status, ○ employment_type, ○ residence, ○ high_glucose, ○ smoking_habit

1.4 Dataset Cleaning

The dataset contained a lot of missing values and duplicates. Initially, we had to drop the duplicate rows which was around 4612 rows leaveing us with 5110 rows.

Next we looked for missing values in the dataset. We found that there were 201 missing values in the `bmi_value` column. Since just dropping these missing values as well would reduce our dataset significantly, we chose to impute it. To maintian a testset that was representative of the original dataset and to prevent data leakage, we performed stratified sampling to split the dataset into training and testing sets prior to any calculations pertaining to the imputation of missing values.

Looking at the distribution of the `bmi_value` column, we found that it was almost normally distributed.

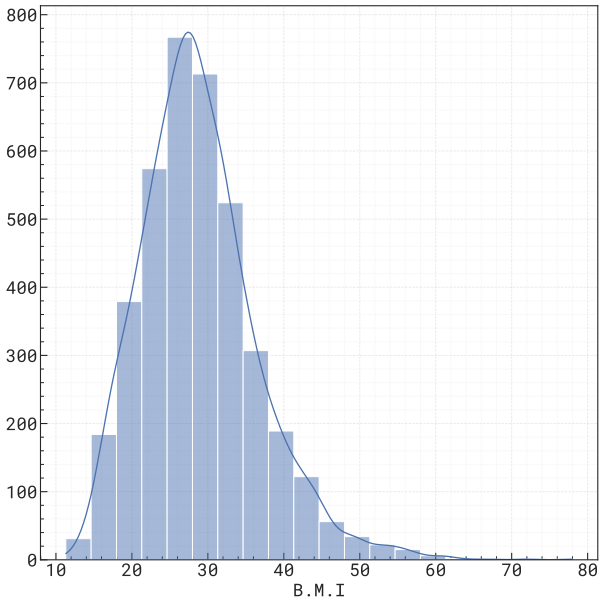

Figure 1.1: BMI Distribution

Therefore, we chose to impute the missing values with the mean of the `bmi_value`. On top of that we noticed that there were no missing values in the `bmi_category` column and that each `bmi_category` was approximately normally distributed. Therefore imputing the missing values with the mean `bmi_value` of each `bmi_category` was a valid approach.

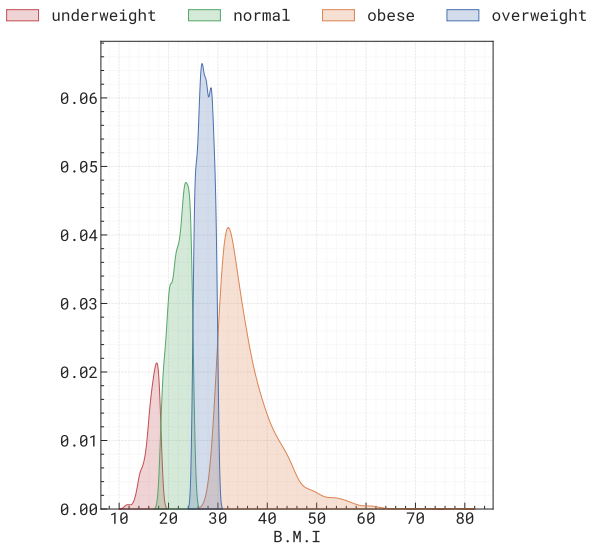

Figure 1.2: BMI KDE plot seperated by categories

Cluster Analysis 2

Since the dataset contained both numeric and categorical type variables, a Principle Component Analysis would be inaccurate. Therefore, we performed a Factor Analysis for Mixed Data.

component	0	1	2	3	4	5	6	7	8	9
variable										
age	0.615015									
has_hypertension										
has_heart_disease										
marital_status										
glucose_level										
bmi_value										
risk_score										
high_glucose										
age_group	0.639189									
bmi_category	0.552375					0.643205				
employment_type	0.657980									
gender										
lifestyle_risk		0.863658	0.589456							
residence							0.614417			
smoking_habit	0.514895	0.884311	0.630753							0.551251

Figure 2.1: FAMD Loadings

By Looking at the loadings we can derive the following factors:

Table 2.1: FAMD Factors

Factor	Variables
Experience	○ age, ○ age_group, ○ employment_type,
Risk Factor	○ smoking_habit, ○ lifestyle_risk

Next by using the K-Means algorithm we were able to identify 4 distinct clusters in the dataset. Averaging with a Silhouette Score of 0.56.

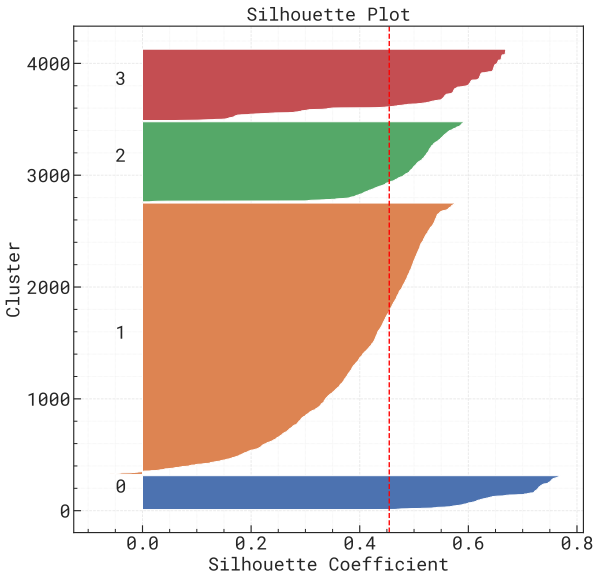

Figure 2.2: Silhouette Score Plot

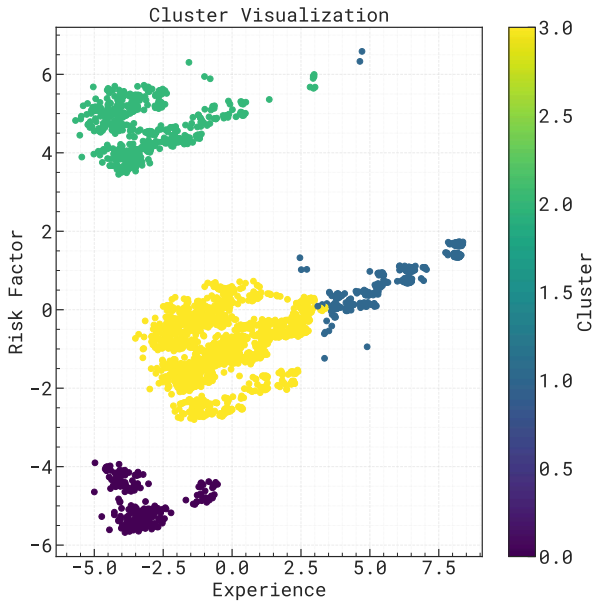

Figure 2.3: Cluster Analysis

Exploratory Data Analysis 3

3.1 Variable distributions

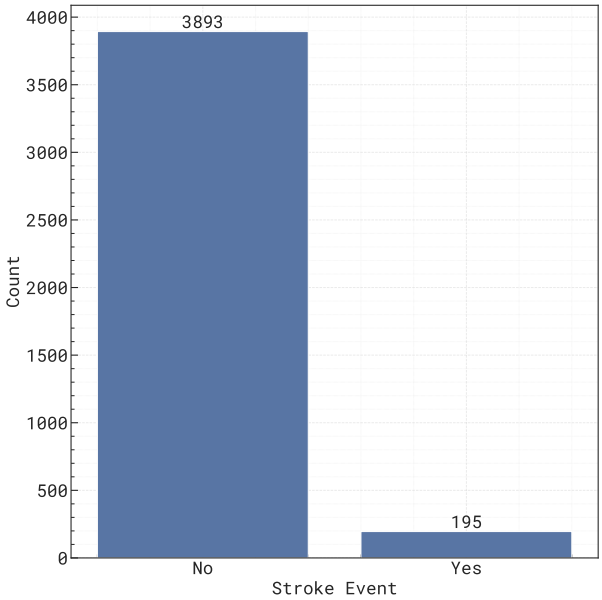

Figure 3.1: Stroke Distribution

This above count plot shows the significant imbalance in the target variable `stroke_event`. Which will be needed to address during the model training phase.

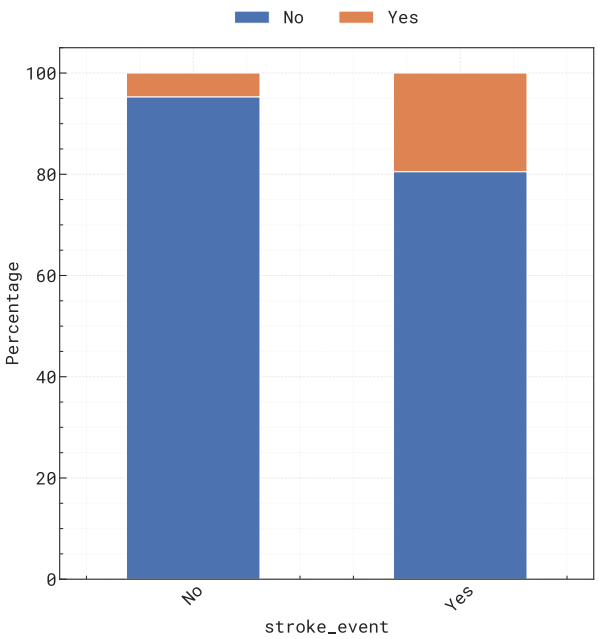

Figure 3.2: Stroke Event vs Has Heart Disease

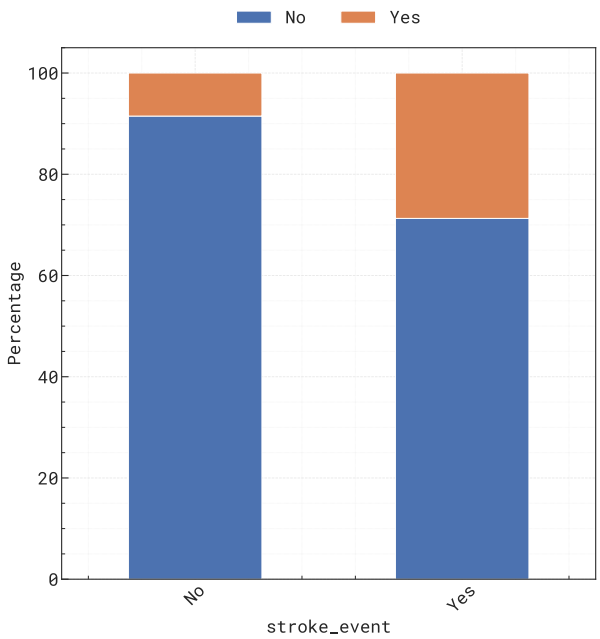

Figure 3.3: Stroke Event vs Has Hypertension

It can be seen that the proportions of the stroke events are significantly higher in the group with heart disease and hypertension. This should be investigated using statistical tests to confirm that this is not some inherent biasdue to the imbalance in the dataset.

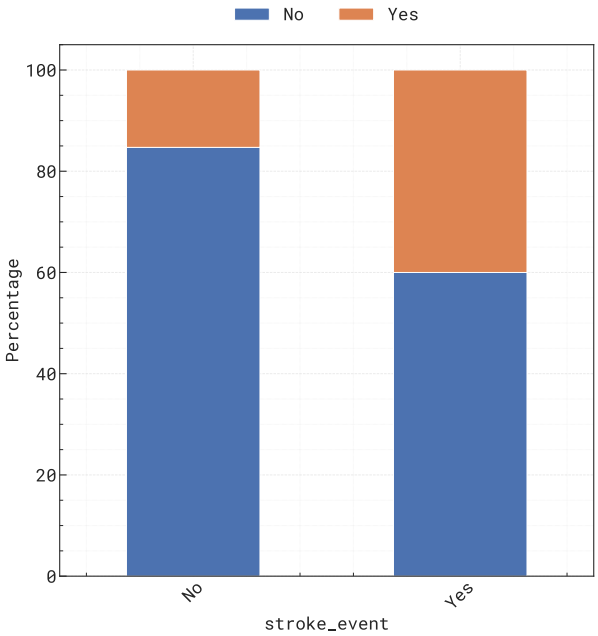

Figure 3.4: Stroke Event vs High Glucose

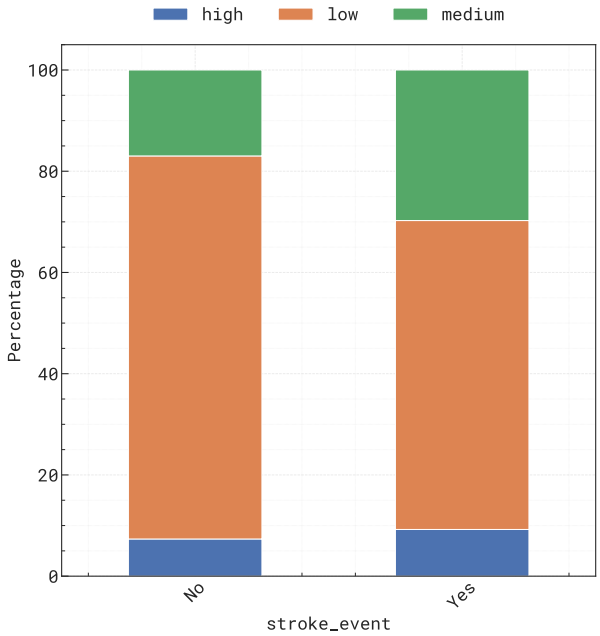

Figure 3.5: Stroke Event vs Lifestyle Risk

3.2 Analytical Approach

The analysis follows a structured framework:

1. Exploratory Data Analysis (EDA): Examination of variable distributions
2. Visual Analysis: Boxplots and bar charts
3. Statistical Testing: Parametric and non-parametric tests

3.3 Variable Classification

Table 3.1: Variable Classification

Category	Variables
Numerical	age, glucose_level, bmi_value, risk_score
Categorical	gender, age_group, has_hypertension, has_heart_disease, marital_status, employment_type, residence, high_glucose, bmi_category, smoking_habit, lifestyle_risk
Ordinal	age_group, bmi_category, lifestyle_risk
Nominal	gender, has_hypertension, has_heart_disease, marital_status, employment_type, residence, high_glucose, smoking_habit

3.4 Statistical Methods

Parametric Tests:

- Chi-Square Test: Assesses association between categorical variables and stroke events
- ANOVA: Compares means of numerical variables across stroke event groups

Non-Parametric Tests:

- Fisher's Exact Test: Alternative to Chi-Square for categorical associations
- Kruskal-Wallis Test: Non-parametric alternative to ANOVA

Significance threshold: alpha = 0.05

3.5 Visualization Style

Visualizations were generated using Matplotlib and Seaborn with custom publication-quality style including consistent font family, grid lines, and clean aesthetics.

Data Analysis 4

4.1 Descriptive Statistics

4.1.1 Numerical Variables

Table 4.1: Summary Statistics for Numerical Variables

Variable	Mean	Std Dev	Min	25%	50%	75%	Max
age	54.73	22.41	0.08	35	55	74	82
glucose_level	127.85	33.68	60	100	118	150	300
bmi_value	27.66	6.11	10	23	27	31	55
risk_score	0.30	0.53	0	0	0	1	2

Age shows mean of approximately 55 years with considerable variation (SD = 22.41). Glucose levels average 127.85 mg/dL with right-skewed distribution. BMI values average 27.66, placing patients in overweight category. Risk score is predominantly low (median = 0).

4.1.2 Categorical Variables

Table 4.2: Distribution of Categorical Variables

Variable	Category	Count	Percentage
gender	F	5,627	57.9%
	M	4,095	42.1%
age_group	senior	4,747	48.8%
	middle	3,418	35.2%
	young	1,557	16.0%
has_hypertension	No	7,945	81.7%
	Yes	1,777	18.3%
has_heart_disease	No	8,597	88.4%
	Yes	1,125	11.6%
marital_status	Married	7,413	76.2%
	Not Married	2,309	23.8%
employment_type	Working	7,693	79.1%
	Government	1,278	13.1%
	Other	751	7.7%
residence	Urban	5,107	52.5%
	Rural	4,615	47.5%
high_glucose	No	7,126	73.3%
	Yes	2,596	26.7%
bmi_category	Obese	4,654	47.9%
	Overweight	2,820	29.0%
	Normal	1,893	19.5%
	Underweight	355	3.7%
smoking_habit	Non-smoker	3,560	36.6%
	Unknown	2,395	24.6%
	Ex-smoker	2,179	22.4%
	Current smoker	1,588	16.3%
lifestyle_risk	Low	6,736	69.3%
	Medium	2,179	22.4%
	High	807	8.3%

Majority are female (57.9%), seniors (48.8%), married (76.2%), working (79.1%). Nearly half are obese (47.9%), 18% have hypertension, 12% have heart disease.

4.2 Visual Analysis

4.2.1 Numerical Variables vs Stroke Event

Box plots reveal:

- Age: Stroke patients show higher median age

- Glucose Level: Higher median in stroke group
- BMI: Similar distributions between groups
- Risk Score: Higher values in stroke group

4.2.2 Categorical Variables vs Stroke Event

Stacked bar charts show:

- Age Group: Senior patients have dramatically higher stroke rates
- Hypertension: Higher stroke rates in hypertensive patients
- Heart Disease: Elevated stroke rates in patients with heart disease
- High Glucose: Increased stroke occurrence with elevated glucose
- BMI Category: Higher BMI shows elevated stroke rates
- Lifestyle Risk: Higher risk correlates with higher stroke rates

4.3 Statistical Testing Results

4.3.1 Parametric Tests

Table 4.3: Chi-Square Test Results

Variable	Chi-Square	DOF	p-value	Significance
age_group	2786.98	2	0.000	Significant
marital_status	745.95	1	0.000	Significant
high_glucose	681.79	1	0.000	Significant
employment_type	638.10	2	0.000	Significant
has_hypertension	572.75	1	0.000	Significant
bmi_category	537.46	3	0.000	Significant
has_heart_disease	445.87	1	0.000	Significant
smoking_habit	294.24	3	0.000	Significant
lifestyle_risk	205.37	2	0.000	Significant
residence	13.97	1	0.000	Significant
gender	2.70	1	0.100	NS

Chi-Square: 10/11 categorical variables significant ($p < 0.05$), only gender not significant.

Table 4.4: ANOVA Results

Variable	F-Statistic	p-value	Significance
age	4906.30	0.000	Significant
risk_score	1025.01	0.000	Significant
glucose_level	682.70	0.000	Significant
bmi_value	NaN	NaN	-

ANOVA: Age, risk_score, glucose_level significant ($p < 0.001$).

4.3.2 Non-Parametric Tests

Table 4.5: Fisher's Exact Test Results

Variable	Odds Ratio	p-value	Significance
marital_status	4.06	0.000	Significant
high_glucose	3.54	0.000	Significant
has_hypertension	3.92	0.000	Significant
has_heart_disease	4.57	0.000	Significant
residence	1.16	0.000	Significant
gender	1.07	0.100	NS

Fisher's Exact: Confirms Chi-Square findings. Heart disease shows highest OR (4.57).

Table 4.6: Kruskal-Wallis Results

Variable	H-Statistic	p-value	Significance
age	3284.18	0.000	Significant
risk_score	978.61	0.000	Significant
glucose_level	366.46	0.000	Significant

Kruskal-Wallis: Confirms ANOVA results.

4.4 Summary of Significant Variables

10 categorical and 3 numerical variables show significant associations with stroke events across both parametric and non-parametric tests.

Discussion, Conclusion and Future Work 5

5.1 Discussion

5.1.1 Key Findings

1. Age is the strongest predictor (Chi Square = 2786.98, F = 4906.30), consistent with known epidemiology where stroke risk doubles each decade after age 55.
2. Cardiovascular comorbidities show strong associations:
   - Hypertension: OR = 3.92 (approximately 4x higher odds)
   - Heart disease: OR = 4.57 (highest odds ratio)
   - High glucose: OR = 3.54
3. Lifestyle factors (smoking, lifestyle risk score) show significant associations with stroke events.
4. Gender shows no significant association, suggesting stroke risk may be more influenced by other factors.

5.1.2 Interpretation

The consistency between parametric and non-parametric tests strengthens validity. Effect sizes provide clinically meaningful insights:

- Heart disease: 4.57x odds
- Hypertension: 3.92x odds
- High glucose: 3.54x odds

The marital status association (OR = 4.06) is likely confounded by age and requires careful interpretation.

5.1.3 Clinical Implications

1. Targeted screening: Prioritize patients with hypertension, heart disease, elevated glucose, and advanced age
2. Preventive interventions: Focus on modifiable risk factors
3. Resource allocation: Direct prevention efforts toward high-risk populations

5.1.4 Limitations

1. Balanced dataset may affect generalizability to real-world populations
2. BMI value shows NaN results indicating data quality issues
3. Cross-sectional design limits causal inference
4. Missing important variables: alcohol consumption, physical activity, family history

5.2 Conclusion

This comprehensive analysis has identified key variables associated with stroke events. Stroke risk is multifactorial, with age, cardiovascular comorbidities, metabolic indicators, and lifestyle factors as primary predictors.

The identification of 10 significant categorical and 3 significant numerical predictors provides a foundation for developing predictive models. Strong statistical significance ($p < 0.001$) suggests these factors should be central to any stroke risk prediction framework.

From a public health perspective, this analysis reinforces:

- Regular blood pressure monitoring and hypertension management
- Glucose level screening and diabetes prevention
- Lifestyle modification programs
- Targeted screening for older adults

5.3 Future Work

1. Predictive model development using machine learning algorithms
2. Validation in independent datasets
3. Investigation of interaction effects between risk factors
4. Improvement of data quality
5. Collection of additional risk factors
6. Development of risk stratification protocols